MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – WEEKLY TEST 01** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 10: Circles** | **MM:** 20

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper physical units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice & Assertion-Reason

MATHS MANTHAN | (5 × 1 = 5 Marks)

Q1. From a point P which is at a distance of 13 cm from the centre O of a circle of radius 5 cm, the length of the tangent drawn to the circle is: [1]

(A) 10 cm (B) 11 cm (C) 12 cm (D) $\sqrt{194}$ cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C) 12 cm

Justification: Since radius $\perp$ tangent at point of contact T, $\triangle OTP$ is right-angled at T. By Pythagoras theorem: $PT = \sqrt{OP^2 - OT^2} = \sqrt{13^2 - 5^2} = \sqrt{169 - 25} = \sqrt{144} = \boxed{\mathbf{12}\text{ cm}}$.

Q2. If two tangents TP and TQ are drawn to a circle with centre O such that $\angle POQ = 110°$, then the measure of $\angle PTQ$ is equal to: [1]

(A) 60° (B) 70° (C) 80° (D) 90°

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) 70°

Justification: In quadrilateral $OPTQ$, $\angle OPT = \angle OQT = 90°$ (radius $\perp$ tangent). By angle sum property, $\angle PTQ + \angle POQ = 180° \implies \angle PTQ = 180° - 110° = \boxed{\mathbf{70°}}$.

Q3. If tangents PA and PB from a point P to a circle with centre O are inclined to each other at an angle of 80°, then $\angle POA$ is: [1]

(A) 50° (B) 60° (C) 70° (D) 80°

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) 50°

Justification: Since OP bisects $\angle APB$, $\angle OPA = \frac{1}{2}(80°) = 40°$. In right $\triangle OAP$ ($\angle OAP = 90°$), $\angle POA = 90° - 40° = \boxed{\mathbf{50°}}$.

Q4. Two concentric circles are of radii 5 cm and 3 cm. The length of the chord of the larger circle which touches the smaller circle is: [1]

(A) 6 cm (B) 7 cm (C) 8 cm (D) 10 cm

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C) 8 cm

Justification: Perpendicular from centre O to the tangent chord AB meets at contact point M ($OM = 3$ cm) and bisects AB. Half-chord $AM = \sqrt{5^2 - 3^2} = 4$ cm, so $AB = 2 \times 4 = \boxed{\mathbf{8}\text{ cm}}$.

Q5. **Assertion (A):** The lengths of two tangents drawn from an external point to a circle are equal.
Reason (R): The tangents drawn from an external point to a circle subtend equal angles at the centre. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B)

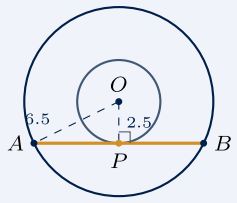
Justification: Both (A) (Theorem 10.2) and (R) are true independently as corresponding parts of congruent right triangles ($\triangle OAP \cong \triangle OBP$ by RHS). However, equal central angles do not serve as the logical cause for equal tangent lengths; hence (R) is not the correct explanation of (A).

SECTION B: Very Short Answer Type — MATHS MANTHAN | (1 × 2 = 2 Marks)

Q6. Two concentric circles are of radii 6.5 cm and 2.5 cm. Find the length of the chord of the larger circle which touches the smaller circle. [2]

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Let O be the common centre, $R = 6.5$ cm, $r = 2.5$ cm, and AB be the chord touching the inner circle at P.



Step 1: Chord Bisection Property [1 Mark]
Since radius $OP \perp AB$ (Theorem 10.1), and the perpendicular from the centre bisects a chord, $AP = PB = \frac{1}{2}AB$.

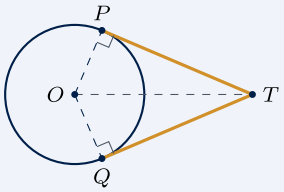
Step 2: Pythagoras Theorem in $\triangle OPA$ [1 Mark]

$$AP = \sqrt{OA^2 - OP^2} = \sqrt{6.5^2 - 2.5^2} = \sqrt{(6.5 - 2.5)(6.5 + 2.5)} = \sqrt{4 \times 9} = 6 \text{ cm}$$

$$\boxed{\text{Length of chord } AB = 2 \times AP = \mathbf{12}\text{ cm}}$$

Q7. Prove that the lengths of tangents drawn from an external point to a circle are equal. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]



Given: A circle with centre O, external point T, and two tangents TP and TQ touching the circle at P and Q.

To Prove: $TP = TQ$.

Construction: Join OP, OQ, and OT. [1 Mark]

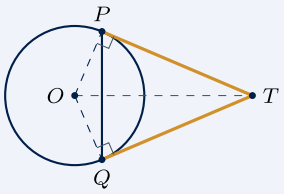
Proof & Deduction: [2 Marks]

1. Since tangent $\perp$ radius at point of contact (Theorem 10.1), $\angle OPT = \angle OQT = 90°$.
2. In right triangles $\triangle OPT$ and $\triangle OQT$:
• $\angle OPT = \angle OQT = 90°$ • $OT = OT$ (Common hypotenuse) • $OP = OQ$ (Radii)
3. By **RHS Congruence Rule**, $\triangle OPT \cong \triangle OQT$.

$$\therefore TP = TQ \quad \text{(by CPCT)} \quad \text{[Proved]}$$

Q8. Two tangents TP and TQ are drawn to a circle with centre O from an external point T. Prove that $\angle PTQ = 2\angle OPQ$. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]



Given: Circle with centre O; tangents TP, TQ from external point T.

To Prove: $\angle PTQ = 2\angle OPQ$. [0.5 Mark]

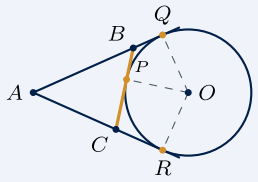
Proof & Deduction: [2.5 Marks]

1. Let $\angle PTQ = \theta$. By Theorem 10.2, $TP = TQ$, so $\triangle TPQ$ is isosceles.
2. Thus, $\angle TPQ = \angle TQP = \frac{1}{2}(180° - \theta) = 90° - \frac{\theta}{2}$.
3. By Theorem 10.1, $OP \perp TP \implies \angle OPT = 90°$.
4. Therefore, $\angle OPQ = \angle OPT - \angle TPQ = 90° - \left(90° - \frac{\theta}{2}\right) = \frac{\theta}{2} = \frac{1}{2}\angle PTQ$.

$$\therefore \angle PTQ = 2\angle OPQ \quad \text{[Proved]}$$

Q9. A circle touches the side BC of $\triangle ABC$ at point P and touches AB and AC produced at Q and R respectively. Prove that $AQ = \frac{1}{2}$(Perimeter of $\triangle ABC$). [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]



Given: Circle touches BC at P, and rays AB, AC produced at Q, R.

To Prove: $AQ = \frac{1}{2}(AB + BC + CA)$. [0.5 Mark]

Proof & Deduction: [2.5 Marks]

1. Tangents from an external point to a circle are equal (Theorem 10.2):
$AQ = AR$, $BQ = BP$, and $CR = CP$.
2. Perimeter$(\triangle ABC) = AB + BC + CA = AB + (BP + PC) + CA$
$= (AB + BQ) + (CR + AC) = AQ + AR = 2AQ$ ($\because AQ = AR$).

$$\therefore AQ = \frac{1}{2}(\text{Perimeter of } \triangle ABC) \quad \text{[Proved]}$$

Q10. Industrial Pulley and Belt Drive Transmission

An engineer designs a mechanical conveyor system where a drive belt runs tautly around a circular drive wheel of radius $r = 5$ cm centred at O. The belt leaves the wheel tangentially at points P and Q and converges at an external tensioner bracket T located 13 cm from the wheel centre ($OT = 13$ cm).

(i) State the mathematical relationship between radius OP and tangent segment TP at the point of contact. [1]

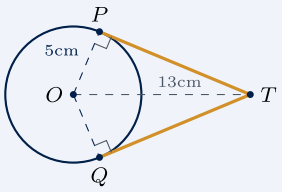
(ii) Calculate the exact length of the tangent belt segment TP. [1]

(iii) Find the measure of central angle $\angle POQ$ if the angle between the belt lines $\angle PTQ = 45°$, OR calculate the area of quadrilateral $OPTQ$. [2]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 4 Marks]

(i) Radius–Tangent Relation: [1 Mark]

Radius is perpendicular to the tangent at the point of contact: $OP \perp TP$, i.e., $\boxed{\angle \mathbf{OPT = 90°}}$.



(ii) Length of Tangent Belt TP: [1 Mark]

In right $\triangle OPT$: $TP = \sqrt{OT^2 - OP^2} = \sqrt{13^2 - 5^2} = \sqrt{144} = \boxed{\mathbf{12}\text{ cm}}$.

(iii) Main Choice — Central Angle $\angle POQ$ (given $\angle PTQ = 45°$): [2 Marks]

In quadrilateral $OPTQ$, $\angle OPT + \angle OQT = 90° + 90° = 180° \implies \angle POQ + \angle PTQ = 180°$.

Therefore, $\angle POQ = 180° - 45° = \boxed{\mathbf{135°}}$.

— OR — (Alternative Choice — Area of Quadrilateral $OPTQ$): [2 Marks]

Area$(\triangle OPT) = \frac{1}{2} \times OP \times TP = \frac{1}{2} \times 5 \times 12 = 30$ cm^2. Since $\triangle OPT \cong \triangle OQT$,

$$\boxed{\text{Area}(OPTQ) = 2 \times 30 = \mathbf{60}\text{ cm}^2}$$

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★